Do People Trust AI More When It Explains Itself?

Design Summary — CEN5728 Advanced HCI, Group 4

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1. The Study

We test whether the **depth** of an AI's explanation changes how much people agree with it, trust it, and catch its mistakes in loan approval. The "AI" is Wizard-of-Oz: every decision is pre-scripted. Each participant reviews 15 loan cases across all three conditions (within-subjects). Six of the cases are deliberately wrong, to test whether explanations help people catch errors or just encourage blind trust (automation bias).

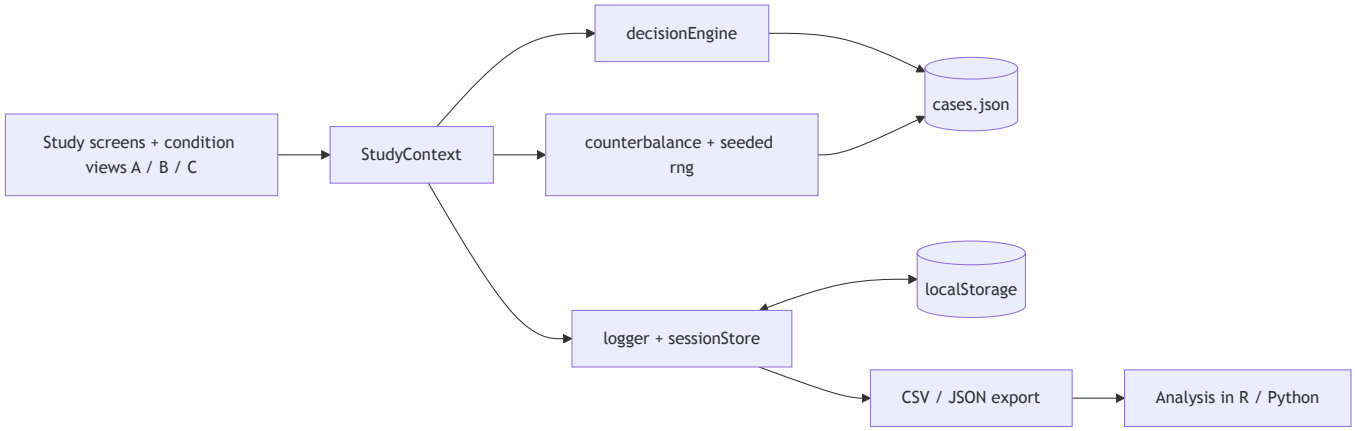
Condition	What the user sees
A Black Box	The decision only: "Approved" / "Rejected"
B Static	The decision plus a fixed written reason
C Interactive	The decision plus a "Why?" button that opens one row per factor (value vs. threshold, pass/fail). Every expand is logged

Lending rules: approve only if income ≥ $45,000, credit history ≥ 3 years, and debt-to-income ≤ 40%. Otherwise reject.

- **H1:** agreement rises with depth, A < B < C.
- **H2:** trust is higher in B and C than in A.
- **H3a:** on wrong cases, agreement is higher in B than in A. **H3b** (exploratory): lower in C than in B.
- **H4:** fairness is highest in C and lowest in A.
- **H5:** decision time is longest in C, and longer times go with better error detection.

2. Architecture

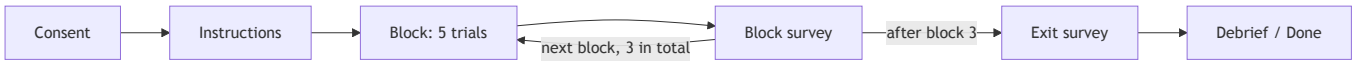
The app is a static React 18 + TypeScript single-page app built with Vite. It has no server: cases come from one JSON case bank (`cases.json`). Responses are saved to `localStorage` after every trial, and the researcher downloads them as CSV/JSON from `/admin` for analysis in R or Python.



- **No backend:** no server to host or keep running, and it works offline on a lab laptop.
- **Seeded PRNG (mulberry32):** it's seeded from the participant number, so every assignment and trial order can be rebuilt exactly and checked for balance.
- **Shared components:** all three views use the same `ApplicantCard`, `DecisionPanel` and rules panel. Only the explanation region differs, so explanation depth is the only thing that changes between conditions.
- **Pure domain layer:** `decisionEngine` and `counterbalance` have no React code, so they can be unit-tested, and the engine checks each case's ground truth against the rules.

3. Participant Flow

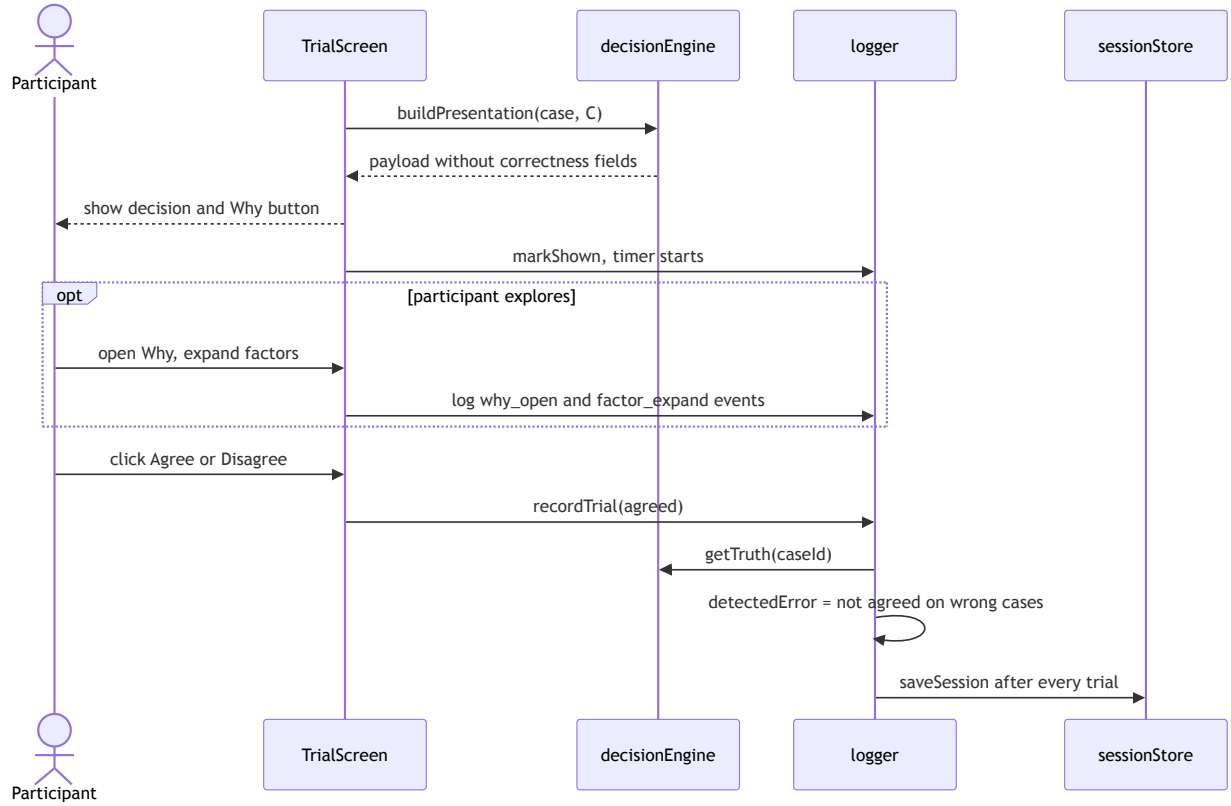
Each session moves through consent, instructions, three blocks of five trials (each followed by a block survey), an exit survey and a debrief.



Counterbalancing. Condition order uses all 6 permutations of A/B/C, and the case sets (S1–S3) are rotated across conditions with a 3×3 Latin square. Together that gives a complete cycle every 18 participants. Each participant does 3 blocks × 5 trials (3 correct and 2 wrong cases per block). Trial order within a block is shuffled with the seeded PRNG, with one rule: the two wrong cases are never both in the first two positions.

4. Key Logic: One Trial

Each trial starts with `decisionEngine.buildPresentation(case, condition)`, which returns the applicant, the AI decision and, in Condition C, the factor breakdown. Correctness fields never reach the view: only `logger` calls `getTruth(caseId)`. The timer runs from when the decision is shown to the Agree/Disagree click, and the session is saved to localStorage after every trial.

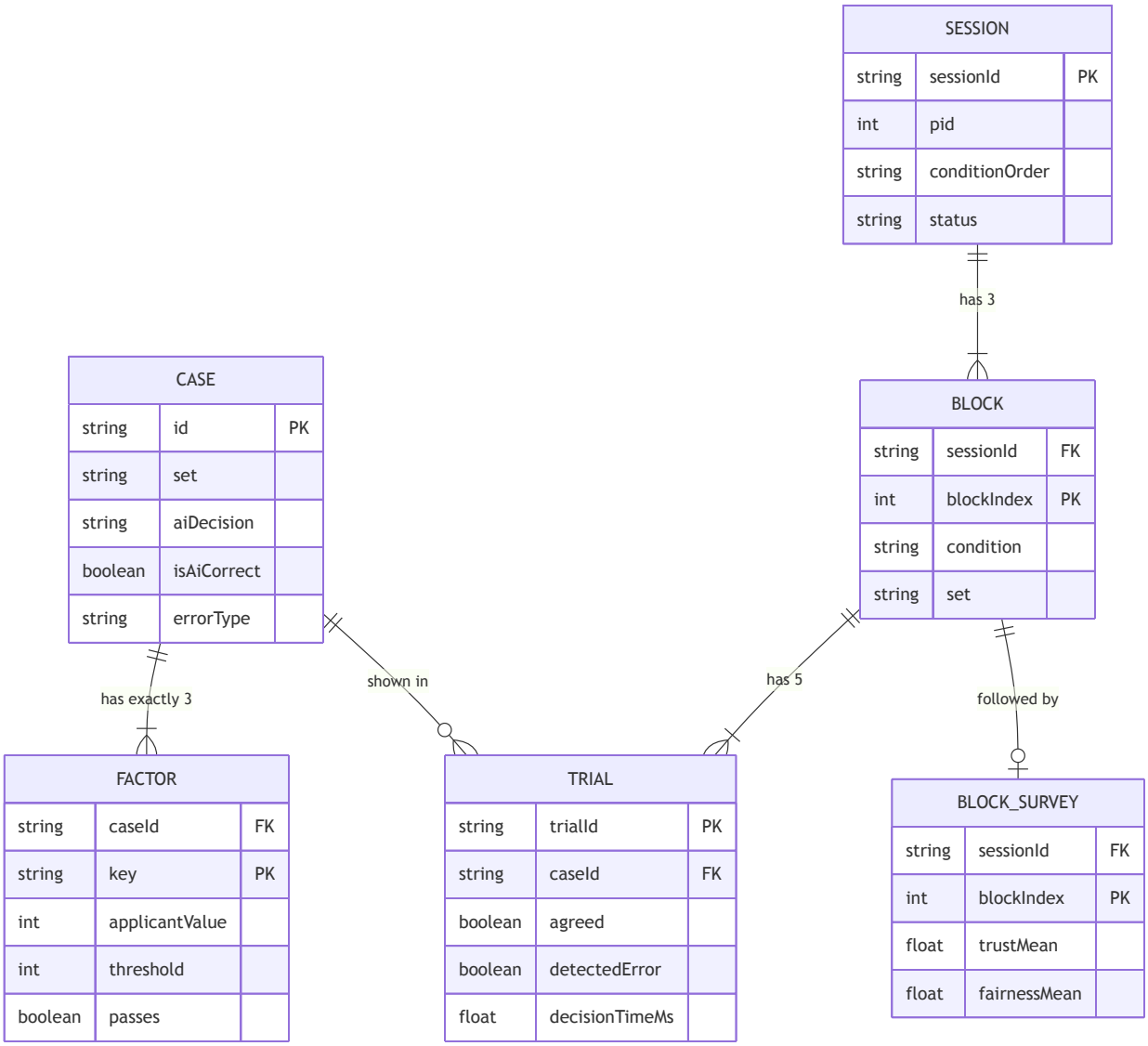


- `detectedError` is `!agreed` on wrong-AI cases and `null` on correct ones.
- `timeToFirstExpandMs` runs from decision shown to the first "Why?" open (C only, `null` in A and B).
- **Hidden-tab time is subtracted:** `decisionTimeMs = decisionTimeRawMs − hiddenMs`.

- **Refresh resumes** the interrupted trial, and its record is flagged `resumed = true`.

5. Data Model

There is no database: the app reads a read-only `cases.json` and keeps one session log in localStorage. The admin page exports it as `trials.csv`, `surveys.csv`, `events.csv` and `sessions.json`.

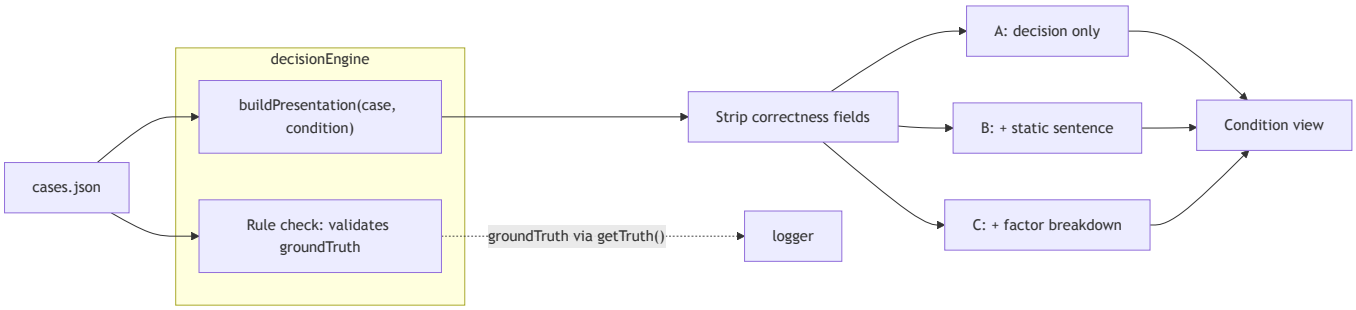


The 15-case bank

id	set	income	history (yrs)	DTI %	AI decision	correct?	error type
case-01	S1	72,000	8	24	APPROVED	Yes	—
case-02	S1	68,000	6	52	APPROVED	**No**	false_approval
case-03	S1	47,500	4	38	REJECTED	**No**	false_rejection
case-04	S1	45,500	4	39	APPROVED	Yes	—
case-05	S1	31,000	5	33	REJECTED	Yes	—
case-06	S2	88,000	12	18	APPROVED	Yes	—
case-07	S2	61,000	2	29	APPROVED	**No**	false_approval
case-08	S2	95,000	10	21	REJECTED	**No**	false_rejection
case-09	S2	51,000	3	40	APPROVED	Yes	—
case-10	S2	54,000	7	58	REJECTED	Yes	—
case-11	S3	79,000	9	27	APPROVED	Yes	—
case-12	S3	43,500	5	31	APPROVED	**No**	false_approval
case-13	S3	76,000	11	16	REJECTED	**No**	false_rejection
case-14	S3	45,000	5	35	APPROVED	Yes	—
case-15	S3	26,000	1	45	REJECTED	Yes	—

6. The Wizard-of-Oz AI

There is no ML model. Every decision is scripted in `cases.json`, and a rule evaluator in `decisionEngine` only computes `groundTruth` to validate the case bank. Correctness fields (`groundTruth`, `isAiCorrect`, `errorType`) are stripped before the payload reaches any view. Each decision appears after a fixed ~800 ms "analyzing" delay in every condition, and no AI confidence is ever shown.



- **B, static sentence:** one plausible sentence of 8–12 words with no numbers, written in the same confident tone for right and wrong decisions, so neither hedging nor wording gives away an error.
- **C, factor breakdown:** rows show the applicant's *true* values against each threshold, so a wrong decision can be checked directly. This is the mechanism behind H3b, which asks whether cheap verification beats a more persuasive explanation.
- **Wrong cases:** 3 false approvals and 3 false rejections. Each set has one clear and one near-threshold wrong case, and `difficulty` is logged as a covariate.

7. Analysis Plan

All analysis runs in R or Python on the exported CSVs. With N = 18, the wrong-case tests have low power, so they are reported with effect sizes and CIs.

Hypothesis	Measure	Primary test
H1: agreement A < B < C	`agreed`, all 15 trials	Mixed-effects logistic regression, random effects for pid and case; planned contrasts
H2: trust B, C > A	Jian trust scale (12 items) per block	Repeated-measures ANOVA (Friedman as fallback), Holm-corrected contrasts
H3a: wrong-case agreement B > A	`agreed` where `isAiCorrect = false`	Logistic mixed model, condition × `isAiCorrect`; contrast B vs A
H3b (exploratory): wrong-case agreement C < B	As H3a, plus `explanationOpened`	Same model; contrast C vs B
H4: fairness C highest, A lowest	Fairness items (2) per block	Repeated-measures ANOVA or Friedman + Page's trend test
H5: time longest in C, and longer time predicts detection	`decisionTimeMs` (visibility-adjusted)	LMM on log decision time; detection ~ time × condition

8. Team & Next Steps

Member	Proposed area
Deepthi Reddy Chelladi	Condition A + shared trial components
Siwani Sah	Condition B + all static explanation strings
Bibek Gupta	Condition C + expand/collapse logging
Afzal Hossain Sohag	Case bank, engine, counterbalancing, tests
Md Shamimul Islam	Study flow, surveys, demo video
Mohamed Sylla	Logging, storage, CSV export, admin, final zip

Open decisions for the team

- **C on wrong cases:** confirm that the breakdown shows true values (recommended).
- **Team split:** the ownership table above is still a proposal.
- **Survey wording:** the fairness items and trust-scale anchor labels are placeholders.

Deliverables: source code (.zip), a demo video of the app running, and a report with screenshots of the code, the app, and group members using it.